

# Beliefs, Expectations, and Job Search

## Behavioral Labor – Week 3

Teaching deck draft

## Central question

- How do beliefs, expectations, perception, and limited information shape job search?
- Week 2 treated worker behavior as a preference problem.
- Week 3 asks when search behavior is instead driven by subjective beliefs about prospects, wages, competition, or fit.
- The discipline remains: object, margin, identifying variation, counterfactual, welfare.

## Bridge from Week 2

| Week 2 object               | Week 3 shift                                        |
|-----------------------------|-----------------------------------------------------|
| Present bias and commitment | Beliefs about returns to search effort              |
| Reference points and effort | Perceived wage offers and reservation behavior      |
| Worker-side margins         | Search intensity, applications, duration, targeting |
| Welfare caution             | Choices may reflect mistaken or sticky expectations |

# Why job search is behaviorally rich

- Search is repeated, high-stakes, and uncertain.
- Feedback is noisy: rejection can mean bad luck, poor targeting, weak skills, or a crowded vacancy.
- Workers may misperceive job-finding probabilities, wage offers, competition, or their own skill fit.
- Firms also lack information about worker skills.
- The result is a two-sided behavioral-information environment.

# Benchmark search problem

$$\max_{s_t, \bar{w}_t} p_t(s_t) \mathbb{E} [V^E(w) \mid w \geq \bar{w}_t] + (1 - p_t(s_t)) V^U - c(s_t)$$

| Object      | Interpretation                                         |
|-------------|--------------------------------------------------------|
| $s_t$       | search effort: time, applications, channels, targeting |
| $\bar{w}_t$ | reservation wage or acceptance threshold               |
| $p_t(s_t)$  | perceived job-finding probability generated by effort  |
| $c(s_t)$    | effort, time, application, and planning costs          |

# Subjective beliefs and search effort

$$\max_{s_t, \bar{w}_t} p_t^b(s_t) \mathbb{E}^b [V^E(w) \mid w \geq \bar{w}_t] + \left(1 - p_t^b(s_t)\right) V^U - c(s_t)$$

- Behavioral wedge can enter through subjective job-finding probabilities.
- It can also enter through perceived wage offers or perceived competition.
- Preferences may be standard while beliefs are biased, sticky, or incomplete.
- The empirical task is to name the belief object and the search margin.

# Unemployment duration and belief updating

- Elicited job-finding beliefs can predict future exits.
- Prediction does not imply correct calibration.
- Long unemployment spells produce repeated but ambiguous signals.
- Weak updating can keep reservation wages high or search strategies unchanged.
- Welfare depends on whether optimism is useful persistence or distorted learning.

# Information provision to job seekers

| Treatment             | Search margin                     | Interpretation issue               |
|-----------------------|-----------------------------------|------------------------------------|
| Brochure or advice    | effort, applications, exits       | information versus motivation      |
| Personalized guidance | targeting and channel choice      | strategy quality versus monitoring |
| Prospect information  | reservation behavior and duration | belief updating versus selection   |

The Altmann-Falk-Jaeger-Zimmermann design is useful because the outcome is search behavior and employment, but the mechanism still needs diagnostic care.



## Two-sided limited information

- Workers lack information about wages, fit, prospects, and sometimes their own skills.
- Firms lack information about worker skills, reliability, and match quality.
- Certification can change worker beliefs and firm screening at the same time.
- The Carranza-Garlick-Orkin-Rankin design is therefore not a pure worker-belief treatment.
- Ask: who learns what, who observes the signal, and which margin moves?

# Wage announcements and perceived competition

- Posted wages raise expected payoffs in the benchmark.
- Workers may also infer that high-wage vacancies are more competitive or selective.
- Applications can respond to the wage as a price and as a signal.
- The Belot-Kircher-Muller design separates wage announcement from hidden job quality concerns.
- The search margin is vacancy choice and application behavior.

# Application costs and selection

- Application costs can be monetary, time-based, cognitive, or psychological.
- Small costs can sharply change who applies.
- The Abebe-Caria-Ortiz-Ospina design links frictions to talent selection.
- Interpretation: liquidity, salience, perceived returns, procrastination, and transport costs can all matter.

# Methods and identification map

| Method                   | Identifies well                  | Main caution                     |
|--------------------------|----------------------------------|----------------------------------|
| Elicited beliefs panel   | reported prospects and updating  | unobserved heterogeneity         |
| Survey expectations      | expected wages and durations     | measurement error                |
| Information experiment   | causal search response           | information versus motivation    |
| Posting experiment       | vacancy and application response | payoff versus competition signal |
| Certification experiment | worker and firm learning         | two-sided treatment              |
| Structural search model  | counterfactuals and welfare      | model dependence                 |

- Reproduce: synthetic Mueller-Spinnewijn-Topa-style belief panel.
- Diagnose: predictive beliefs, duration dependence, weak updating, and reservation behavior.
- Transfer: synthetic certification design inspired by Carranza-Garlick-Orkin-Rankin.
- Optional extension: classify wage announcements as payoff signals and perceived-competition signals.

- Measuring beliefs is not enough; designs must separate beliefs from constraints and selection.
- Information provision can change beliefs, attention, motivation, and compliance at once.
- Two-sided information links behavioral job search to hiring and firm screening.
- Welfare depends on whether interventions improve choices, reallocate applications, or shift pressure.

- Week 3: beliefs, expectations, perception, and information in job search.
- Week 4: attention, salience, complexity, and policy design.
- The common object is perceived choice environment.
- The next question is how workers respond when the rule exists but is hard to notice, understand, or act on.

- Behavioral job search is not one mechanism.
- Beliefs can be predictive and still biased or weakly updated.
- Wage postings, certifications, and application costs are search-information objects.
- Empirical claims must identify the belief object, search margin, and standard alternatives.